

Tien Dung Vu

AI Engineer | Data Scientist / Analyst

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist focused on taking machine learning from notebook to production: NLP, computer vision, and LLM-powered agentic systems, grounded in rigorous data analysis and statistical modelling with Python and SQL. As an international candidate, earned readiness the hard way, completing AI engineering internships across two countries (Vietnam and Australia) and building production-grade systems to match and exceed domestic peers. Comfortable across the full lifecycle: ETL and data pipelines, model training and evaluation, API-backed deployment with Docker, and stakeholder-ready dashboards and reporting.

CORE COMPETENCIES

Machine Learning & NLP

Computer Vision (CNNs)

LLM & Agentic Systems (LangGraph, RAG)

Data Analysis & Statistical Modelling

Python & SQL

ETL & Data Pipelines

Model Evaluation & Validation

Dashboards & Reporting (Looker Studio)

WORK EXPERIENCE

Affinity Marketing Australia May 2026 - Present

AI-Driven Marketing Intern - Sydney, Australia

- Built 4 end-to-end content automation pipelines using Claude Code and n8n (SEO blog expander, video-to-content-pack, blog-to-social repurposing, ad copy refresh), eliminating manual production steps across recurring campaign cycles
- Developed a Google Ads copy generator producing 15+ Claude-generated variants and an email subject-line A/B testing pipeline, reducing copy turnaround for paid and organic channels
- Built a Looker Studio cross-channel dashboard consolidating Google Analytics 4 and Search Console data into a single performance view with automated weekly stakeholder summaries
- Produced process flow documentation and business requirements documents (BRDs) for automation initiatives, plus an AI automation playbook covering all pipelines and tools for team handover

CMC Corporation Oct 2024 - Apr 2025

AI Engineer Intern - Vietnam

- Built computer vision systems for image classification using Convolutional Neural Networks (CNNs)
- Developed end-to-end deep learning pipelines covering data preparation, training, evaluation, and hyperparameter tuning
- Applied data augmentation and normalisation techniques to improve model generalisation, iteratively refining architectures to optimise accuracy

FPT Vietnam Sep 2023 - Feb 2024

AI Engineer Intern - Vietnam

- Applied supervised learning algorithms (Linear Regression, Logistic Regression, Decision Trees) on structured datasets and K-Means clustering for unsupervised pattern discovery
- Developed NLP classification models using word embeddings and text-processing techniques
- Conducted data analysis, visualisation, and experimentation in Google Colab, interpreting model results to support decision-making

OACIS - Sydney Airport Oct 2025 - Present

Passenger Service Agent - Sydney, Australia

- Managed passenger flow and resolved enquiries in a fast-paced, safety-focused environment, building strong communication and stakeholder management skills across airline staff, security, and operations teams

PROJECTS

Pacific Wings Airline Strategy Simulator

Python, FastAPI, Next.js, TypeScript, XGBoost, scikit-learn, LangGraph, Gemini AI, PostgreSQL, Docker

- Built a full-stack airline analytics and strategy platform grounded in real aviation, macroeconomic, and cost data, with an ETL pipeline ingesting BITRE aviation statistics, World Bank indicators, and real competitor schedules, fares, and ratings
- Trained and validated an XGBoost demand forecasting model ($R^2=0.952$, MAPE=15.3% on a held-out 2024 test set) with 5-fold cross-validation, prediction intervals, and a custom explainable 0-100 confidence-scoring system
- Built a what-if simulation engine and Monte Carlo risk simulator sampling real fuel-price and GDP-growth volatility to produce profit, passenger, and load-factor probability distributions for any route or scenario
- Implemented a LangGraph five-agent strategy pipeline and a conversational Gemini-backed executive chat agent with 11 function-calling tools spanning route simulation, forecasting, P&L analysis, and network ranking
- Led a full data-integrity audit across the 11-page Next.js dashboard, tracing every displayed metric to its source and removing ~30 fabricated or mislabeled UI elements

Research LLM Agent (CLI)

Python, LangGraph, Google Gemini, SerpAPI, Docker

- Designed a 4-node LangGraph pipeline (GenerateQueries, WebSearch, Reflect, Synthesize) with a conditional reflection loop that routes back to web search when coverage gaps are detected, enabling multi-round research on a single query
- Implemented parallel SerpAPI search with URL deduplication, cited answer synthesis, and layered error handling; wrote a 5-scenario test suite (including HTTP 429 throttling and timeouts) and containerised with Docker

Australian Media Sentiment and Framing of AI

Python, VADER, RoBERTa, FinBERT, BERTopic, Guardian API | University of Technology Sydney

- Built a Python ETL pipeline collecting 2,622 Guardian Australia articles via API with rate-limited pagination and multi-criteria relevance filtering, refining the corpus to 1,033 AI-related articles spanning 2017-2026
- Applied a three-model sentiment framework (VADER, RoBERTa, FinBERT) with a sliding-window approach for full article bodies, detecting a persistent headline-body sentiment divergence across all 10 major Guardian sections
- Quantified inter-model agreement, applied BERTopic to identify dominant risk themes, and delivered 6 evidence-based editorial recommendations to The Sydney Morning Herald on AI coverage positioning

Misinformation Detection System

Python, scikit-learn, TF-IDF, Naive Bayes, NLP | University of Technology Sydney

- Led a team of 5 to build a binary fake-news classifier: full NLP preprocessing pipeline with parallelised text cleaning, TF-IDF vectorisation (40,000 features), and a Multinomial Naive Bayes classifier evaluated with precision, recall, F1, and confusion matrix analysis
- Persisted trained model and vectoriser for reuse and shipped an interactive CLI loop for real-time classification of unseen articles

EDUCATION

Bachelor of Artificial Intelligence, University of Technology Sydney (UTS)

Graded: High Distinction

SKILLS

Programming: Python, SQL, Java, Swift, SwiftUI

Machine Learning & AI: NLP, Computer Vision, CNNs, BERT, TF-IDF, Naive Bayes, Text Classification, Sentiment Analysis, Topic Modelling

LLM Systems: LangGraph, LLM Pipeline Design, Agentic Workflow Design, Prompt Engineering, RAG, Information Retrieval

Data & Analytics: Data Analysis, Data Cleaning, Feature Engineering, Exploratory Data Analysis, Statistical Analysis

Software Engineering: API Integration, Backend Development, CLI Applications, Object-Oriented Programming, System Design

Tools & Platforms: Google Colab, Docker, Git, GitHub, REST APIs, Pytest, HubSpot, Salesforce, Looker Studio, n8n

Collaboration: Team Leadership, Agile Development, Project Delivery, Technical Documentation, Cross-functional Communication